School of Computer Science Engineering and Information Systems

Winter Semester 2025-26

M. Tech Integrated Software Engineering

ISWE406P-Agile Development Process and DevOps Lab

Final Assessment Test

System: Smart Electricity Billing System

Set-6

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<u>Problem Statement:</u>

Ensure electricity usage is calculated and billed correctly

Tasks:

1.Biling logic

2.Junit tests for tariff

3.Implement CI/CD pipeline

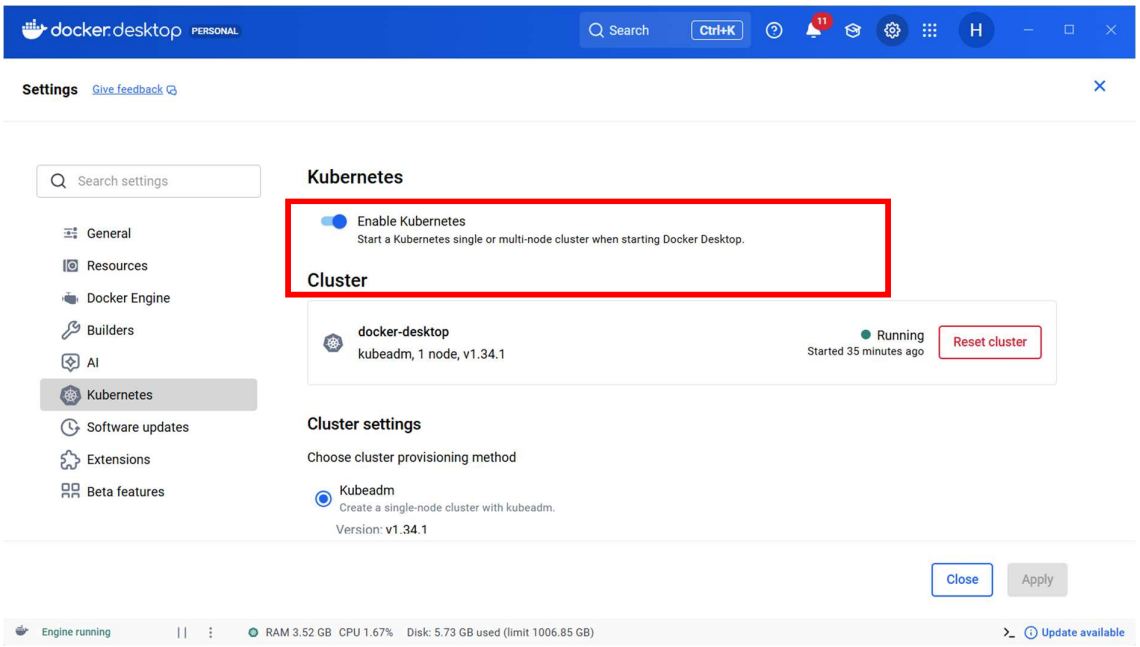
4.Deploy on Docker and kubernetes

Step1

Install Docker Desktop

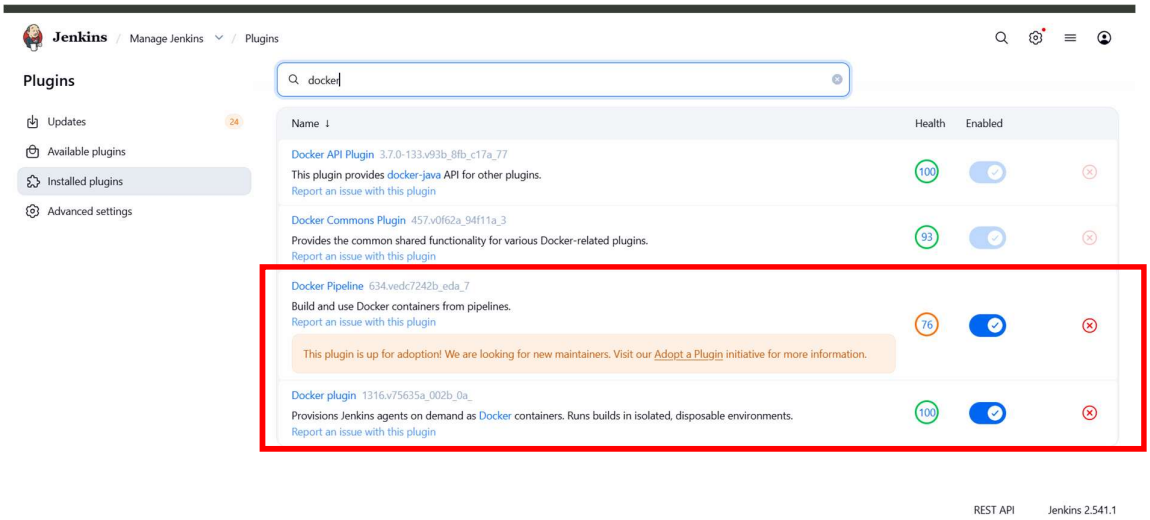
Step2

Enable Kubernetes



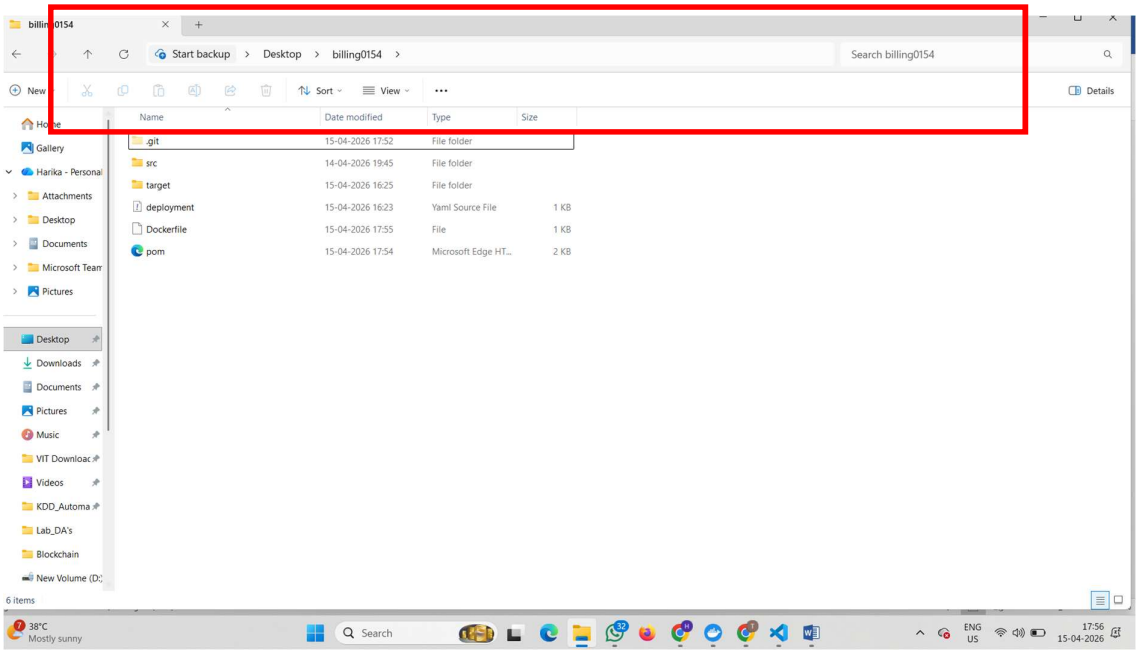
Step 3

In Jenkin, enable two plugins (Docker and Docker Pipeline) and Install them



Step 4 - Create Simple Java Application

Folder: billing0154(Electricity billing Application) and structure of my project folder



Task-1) Billing Calculation Logic [Java program Development]

Java file: **BillingCalculator.java**

```
package com.electricity;
public class BillingCalculator {
    private static final int[] SLAB_LIMITS = {100, 200, 500,
Integer.MAX_VALUE};
    private static final double[] SLAB_RATES = {3.00, 5.00, 7.00, 9.00}; // in
rupees
    private static final double FIXED_CHARGE = 50.00

    /**
     * Calculate total bill amount based on units consumed.
     * @param unitsConsumed number of units (kWh)
     * @return total bill amount in rupees (rounded to 2 decimals)
     * @throws IllegalArgumentException if unitsConsumed < 0
     */
    public static double calculateBill(double unitsConsumed) {
        if (unitsConsumed < 0) {
            throw new IllegalArgumentException("Units consumed cannot be
negative");
        }

        double energyCharge = 0.0;
```

```java
        double remaining = unitsConsumed;
        double prevLimit = 0;

        for (int i = 0; i < SLAB_LIMITS.length; i++) {
            int limit = SLAB_LIMITS[i];
            double slabUnits = Math.min(remaining, limit - prevLimit);
            if (slabUnits > 0) {
                energyCharge += slabUnits * SLAB_RATES[i];
                remaining -= slabUnits;
            }
            prevLimit = limit;
            if (remaining <= 0) break;
        }

        double total = energyCharge + FIXED_CHARGE;
        // Round to 2 decimal places
        return Math.round(total * 100.0) / 100.0;
    }

    /**
     * Get detailed breakdown of bill calculation.
     * @param unitsConsumed units
     * @return formatted string with slab details and total
     */
    public static String getBillBreakdown(double unitsConsumed) {
        if (unitsConsumed < 0) {
            return "Invalid units";
        }
        StringBuilder sb = new StringBuilder();
        sb.append(String.format("Units Consumed: %.2f kWh\n", unitsConsumed));
        sb.append(String.format("Fixed Charge: ₹%.2f\n", FIXED_CHARGE));

        double energyCharge = 0.0;
        double remaining = unitsConsumed;
        double prevLimit = 0;
        sb.append("Slab Details:\n");

        for (int i = 0; i < SLAB_LIMITS.length; i++) {
            int limit = SLAB_LIMITS[i];
            double slabUnits = Math.min(remaining, limit - prevLimit);
            if (slabUnits > 0) {
                double slabCost = slabUnits * SLAB_RATES[i];
                energyCharge += slabCost;
                sb.append(String.format("  %.2f units @ ₹%.2f = ₹%.2f\n",
slabUnits, SLAB_RATES[i], slabCost));
                remaining -= slabUnits;
            }
            prevLimit = limit;
```

```
            if (remaining <= 0) break;
        }

        double total = energyCharge + FIXED_CHARGE;
        sb.append(String.format("Energy Charge: ₹%.2f\n", energyCharge));
        sb.append(String.format("Total Bill: ₹%.2f\n", Math.round(total *
100.0) / 100.0));
        return sb.toString();
    }

    public static void main(String[] args) {
        // Demo
        double units = 350;
        System.out.println(getBillBreakdown(units));
        System.out.println("Total: ₹" + calculateBill(units));
    }
}
```

Task-2) JUnit tests for tariff calculations

<u>Test file</u>: **BillingCalculatorTest.java**

```
package com.electricity;

import static org.junit.jupiter.api.Assertions.assertEquals;
import static org.junit.jupiter.api.Assertions.assertThrows;
import static org.junit.jupiter.api.Assertions.assertTrue;
import org.junit.jupiter.api.DisplayName;
import org.junit.jupiter.api.Test;
import org.junit.jupiter.params.ParameterizedTest;
import org.junit.jupiter.params.provider.CsvSource;

class BillingCalculatorTest {

    @Test
    @DisplayName("Zero units should incur only fixed charge")
    void testZeroUnits() {
        assertEquals(50.00, BillingCalculator.calculateBill(0));
    }

    @Test
    @DisplayName("Negative units should throw exception")
    void testNegativeUnits() {
        assertThrows(IllegalArgumentException.class, () ->
BillingCalculator.calculateBill(-10));
    }

    @ParameterizedTest
```

```java
    @CsvSource({
        "50, 200.00",
        "100, 350.00",
        "150, 600.00",
        "200, 850.00",
        "250, 1200.00",
        "350, 1900.00",
        "600, 3850.00"
    })
    void testBillCalculation(double units, double expected) {
        double actual = BillingCalculator.calculateBill(units);
        assertEquals(expected, actual, 0.01);
    }

    @Test
    @DisplayName("Check exact slab boundaries")
    void testSlabBoundaries() {
        assertEquals(350.00, BillingCalculator.calculateBill(100));
        assertEquals(355.00, BillingCalculator.calculateBill(101));
        assertEquals(850.00, BillingCalculator.calculateBill(200));
        assertEquals(857.00, BillingCalculator.calculateBill(201));
    }

    @Test
    @DisplayName("Breakdown string contains expected text")
    void testGetBillBreakdown() {
        String breakdown = BillingCalculator.getBillBreakdown(150);
        assertTrue(breakdown.contains("Units Consumed: 150.00 kWh"));
        assertTrue(breakdown.contains("Fixed Charge: ₹50.00"));
        assertTrue(breakdown.contains("Total Bill: ₹600.00"));
    }
}
```

Output: [Junit Testing]

```
PS C:\Users\Lenovo\Desktop\billing0154\src\main\java> java com.electricity.BillingCalculator
>>
Units Consumed: 350.00 kWh
Fixed Charge: ?50.00
Slab Details:
>>
Units Consumed: 350.00 kWh
Fixed Charge: ?50.00
Slab Details:
Units Consumed: 350.00 kWh
Fixed Charge: ?50.00
Slab Details:
Fixed Charge: ?50.00
Slab Details:
Slab Details:
  100.00 units @ ?3.00 = ?300.00
  100.00 units @ ?5.00 = ?500.00
  150.00 units @ ?7.00 = ?1050.00
Energy Charge: ?1850.00
Total Bill: ?1900.00

Total: ?1900.0
```

Maven File:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.electricity</groupId>
    <artifactId>electricity-billing</artifactId>
    <version>1.0.0</version>
    <packaging>jar</packaging>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>5.10.0</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.11.0</version>
            </plugin>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.0.0-M9</version>
            </plugin>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.2.0</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>com.electricity.BillingCalculator</main
Class>
```

```
                        </manifest>
                    </archive>
                </configuration>
            </plugin>
        </plugins>
    </build>
</project>
```

Step-5

Local building

Result:

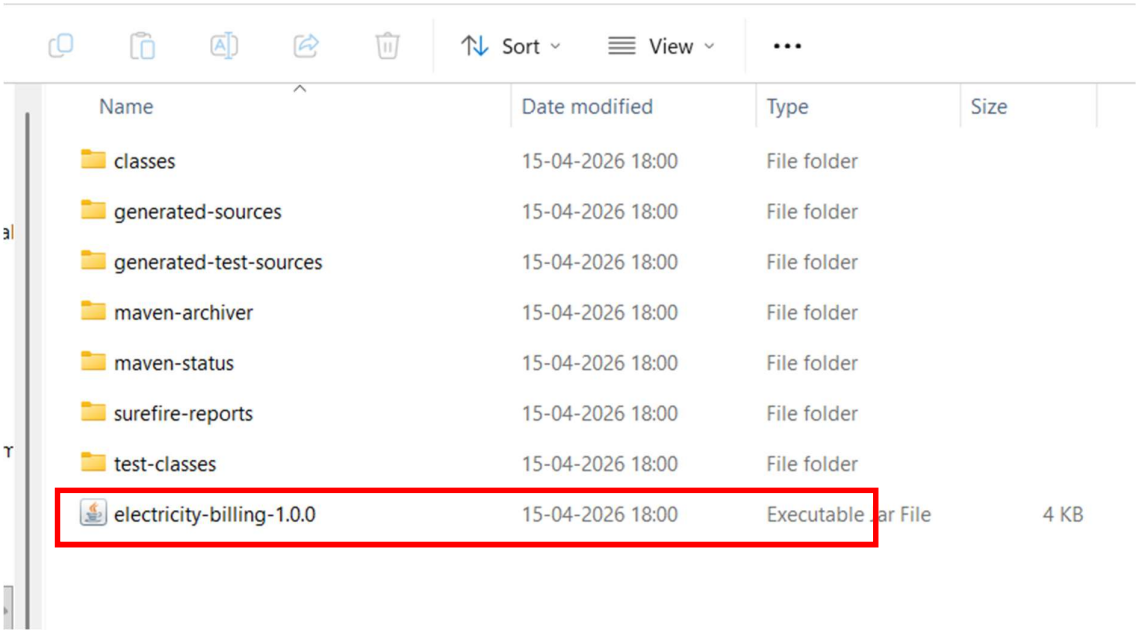
mvn clean test

```
[INFO] Results:
[INFO]
[INFO] Tests run: 11, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  7.838 s
[INFO] Finished at: 2026-04-15T17:59:14+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Lenovo\Desktop\billing0154>
```

mvn clean package

```
PS C:\Users\Lenovo\Desktop\billing0154> mvn clean package
[INFO] Tests run: 11, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.2.0:jar (default-jar) @ electricity-billing ---
[INFO] Building jar: C:\Users\Lenovo\Desktop\billing0154\target\electricity-billing-1.0.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  12.958 s
[INFO] Finished at: 2026-04-15T18:00:12+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Lenovo\Desktop\billing0154>
```

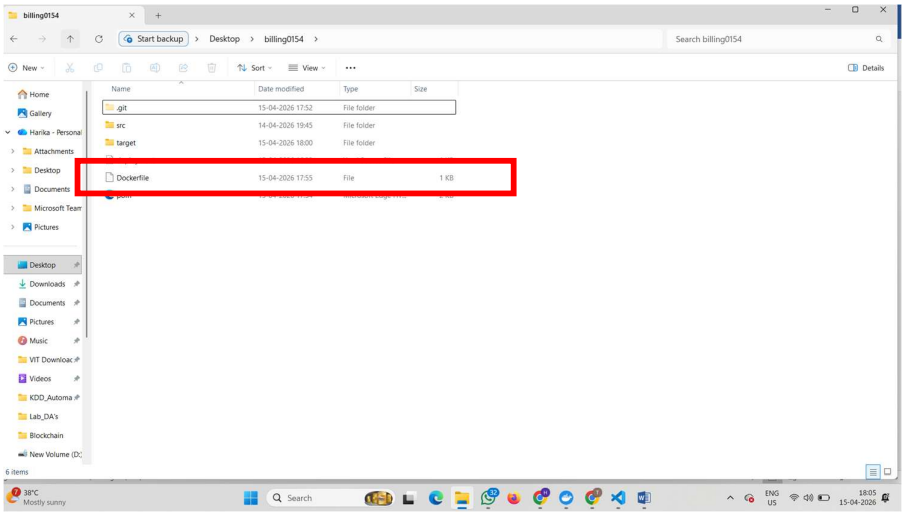
It generates: target/electricity-billing-1.0.0.jar



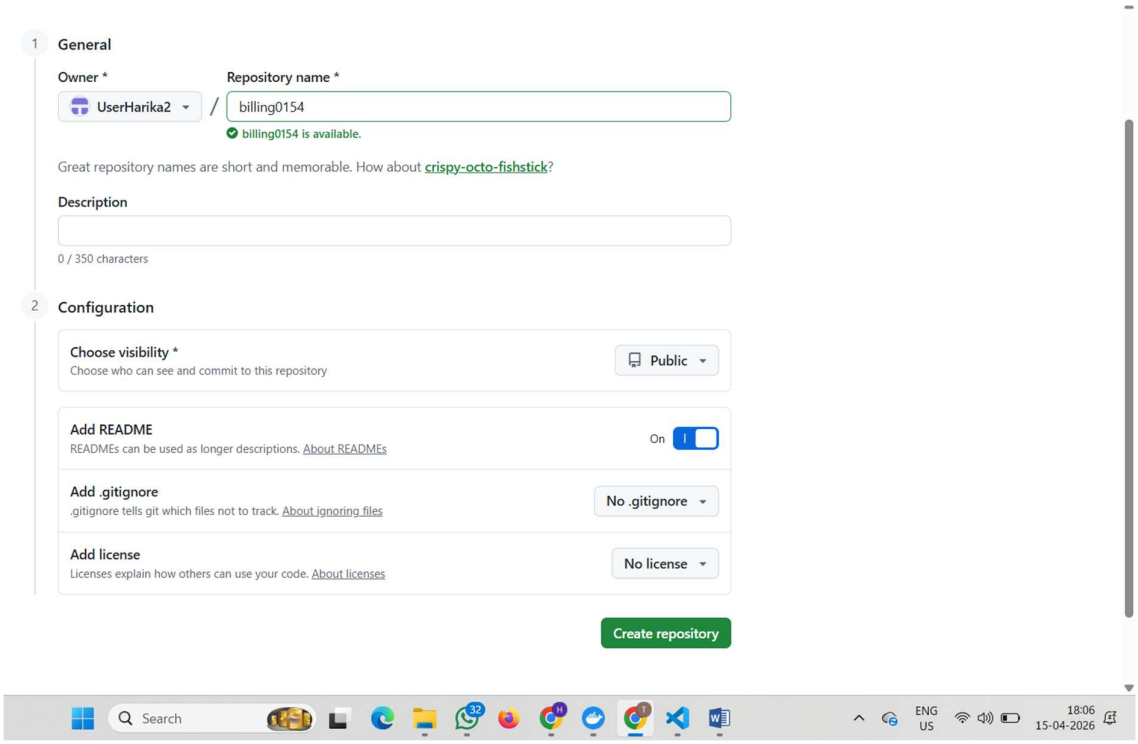
Name	Date modified	Type	Size
classes	15-04-2026 18:00	File folder	
generated-sources	15-04-2026 18:00	File folder	
generated-test-sources	15-04-2026 18:00	File folder	
maven-archiver	15-04-2026 18:00	File folder	
maven-status	15-04-2026 18:00	File folder	
surefire-reports	15-04-2026 18:00	File folder	
test-classes	15-04-2026 18:00	File folder	
electricity-billing-1.0.0	15-04-2026 18:00	Executable Jar File	4 KB

<u>Create Dockerfile</u> Dockerfile

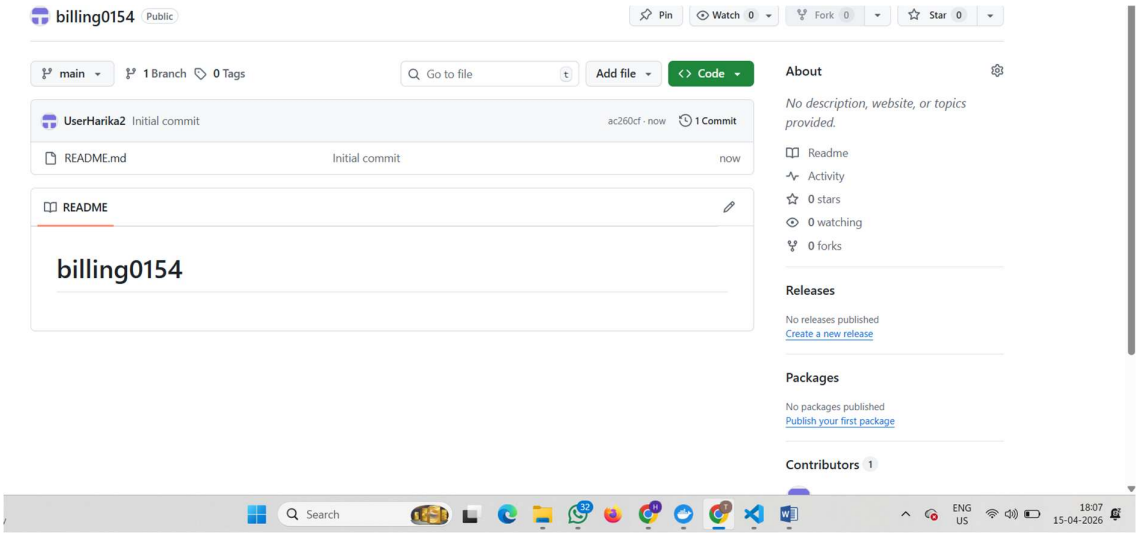
```
FROM eclipse-temurin:17-jdk-alpine
WORKDIR /app
COPY target/electricity-billing-1.0.0.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```



Step 6 -create a repo and push code to github:



Initial phase:



git init

git add .

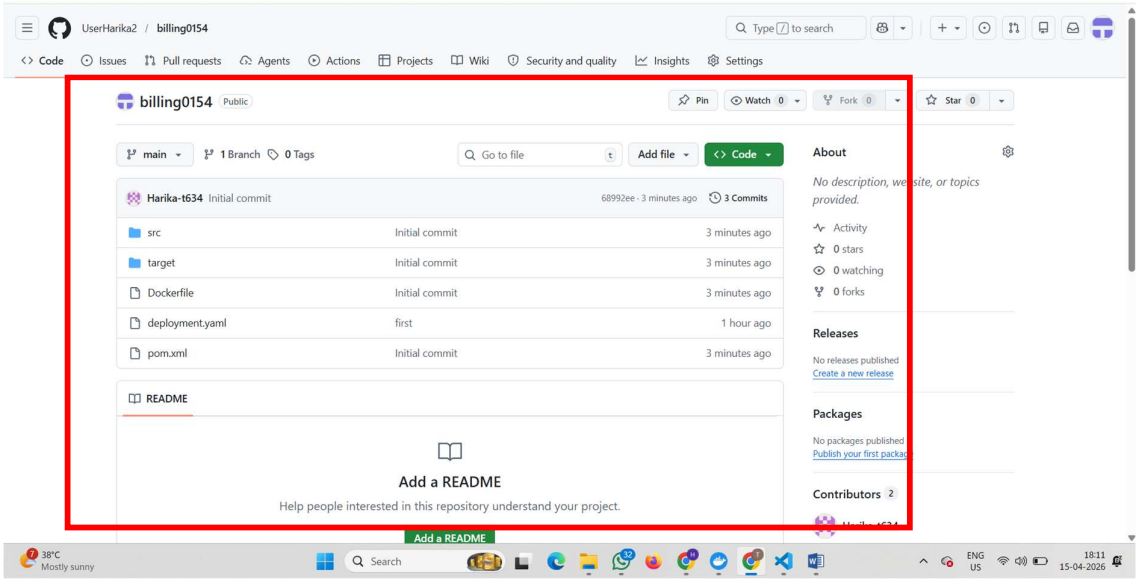
git commit -m "Initial commit to git hub”

git branch -M main

git remote add origin https://github.com/UserHarika2/billing0154.git

git push origin main

```
PS C:\Users\Lenovo\Desktop\billing0154> git init
Reinitialized existing Git repository in C:/Users/Lenovo/Desktop/billing0154/.git/
PS C:\Users\Lenovo\Desktop\billing0154> git add .
PS C:\Users\Lenovo\Desktop\billing0154> git commit -m "Initial commit"
On branch main
nothing to commit, working tree clean
PS C:\Users\Lenovo\Desktop\billing0154> git push origin main
Everything up-to-date
PS C:\Users\Lenovo\Desktop\billing0154> git remote set-url origin https://github.com/UserHarika2/billing0154.git
PS C:\Users\Lenovo\Desktop\billing0154> git push origin main
To https://github.com/UserHarika2/billing0154.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/UserHarika2/billing0154.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
PS C:\Users\Lenovo\Desktop\billing0154> git push origin main --force
Enumerating objects: 86, done.
Counting objects: 100% (86/86), done.
Delta compression using up to 12 threads
Compressing objects: 100% (57/57), done.
Writing objects: 100% (86/86), 62.74 KiB | 1.65 MiB/s, done.
Total 86 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
To https://github.com/UserHarika2/billing0154.git
```



Step 7

Create Deployment file for jenkins and Docker and push it into github

```
PS C:\Users\Lenovo\Desktop\billing0154> git add .
PS C:\Users\Lenovo\Desktop\billing0154> git commit -m "Deployment commit"
[main 2bf849e] Deployment commit
 1 file changed, 11 insertions(+), 11 deletions(-)
PS C:\Users\Lenovo\Desktop\billing0154> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 449 bytes | 449.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/UserHarika2/billing0154.git
   68992ee..2bf849e  main -> main
PS C:\Users\Lenovo\Desktop\billing0154>
```

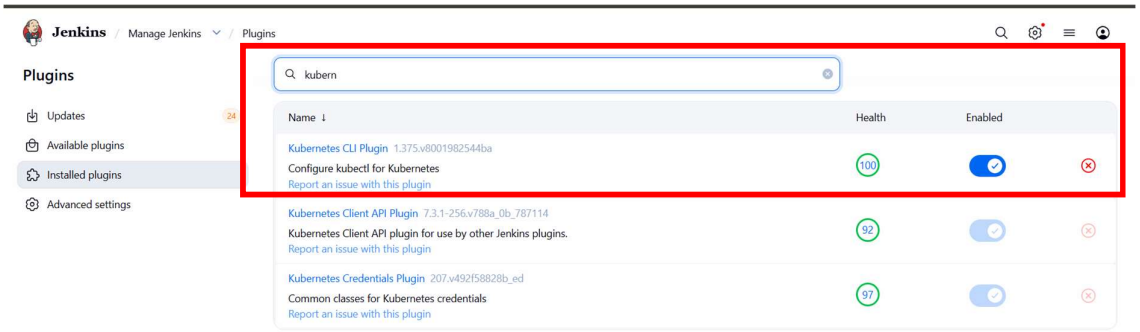
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: electricity-billing-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: electricity-billing
  template:
    metadata:
      labels:
        app: electricity-billing
    spec:
      containers:
      - name: billing-app
        image: your-dockerhub/electricity-billing:latest
        ports:
        - containerPort: 8080   # Not a web server, but we keep for
consistency
        resources:
          requests:
            memory: "128Mi"
            cpu: "100m"
---
apiVersion: v1
kind: Service
metadata:
```

```
  name: electricity-billing-service
spec:
  type: NodePort
  selector:
    app: electricity-billing
  ports:
    - port: 8080
      targetPort: 8080
      nodePort: 30083
```

Step 8

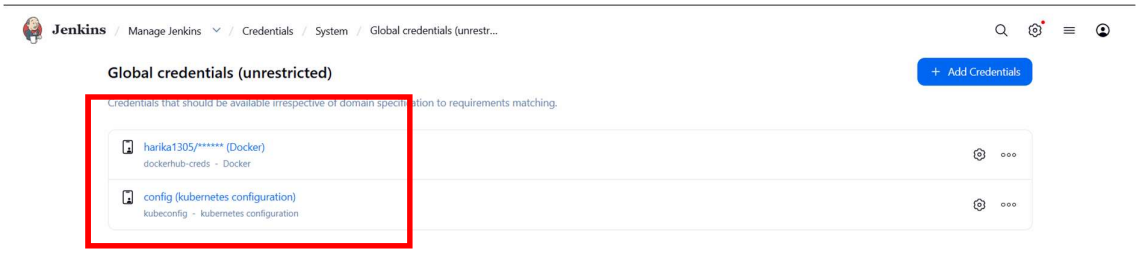
Configure Jenkins

Installed Kubernetes CLI plugin

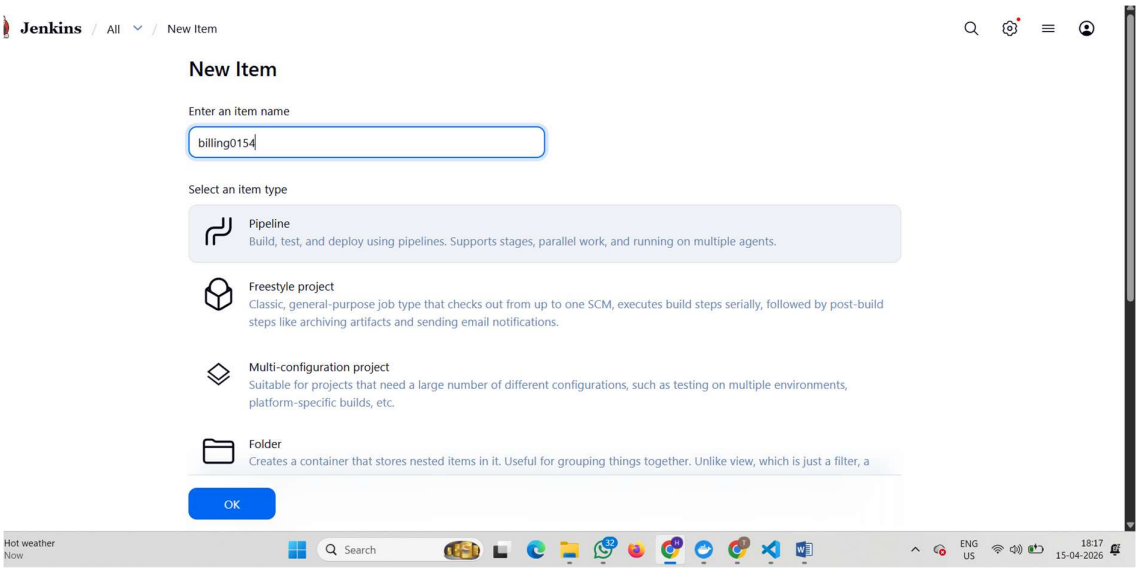


Step 9

Add DockerHub Credentials and Kube config



<u>Step 10</u> - Create Jenkins Pipeline Job **CI/CD pipeline**

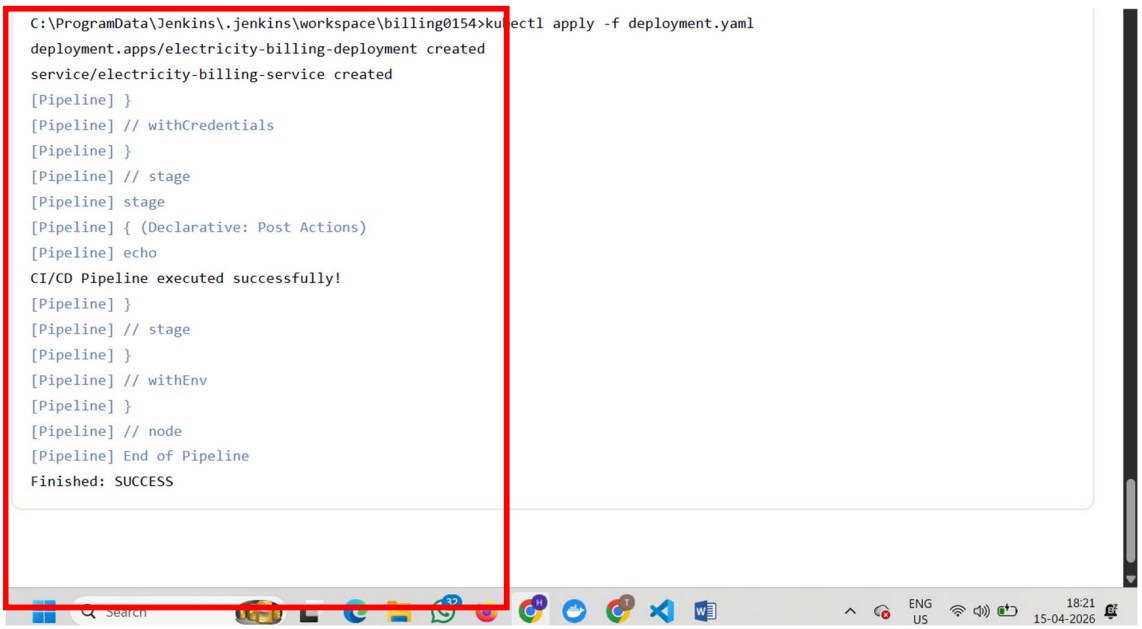


Task-3) CI/CD pipeline create and execution

```
pipeline {
  agent any
  environment {
    DOCKER_IMAGE = "harika1305/electicity-billing:latest"
  }
  stages {
    stage('Clean Workspace') {
      steps {
        deleteDir()
      }
    }
    stage('Clone Repository') {
      steps {
        git branch: 'main',
```

```
        url: 'https://github.com/UserHarika2/billing0154.git'
    }
}
stage('Build with Maven') {
    steps {
        bat 'mvn clean package'
    }
}
stage('Build Docker Image') {
    steps {
        bat "docker build -t %DOCKER_IMAGE% ."
    }
}
stage('Push to DockerHub') {
    steps {
        withCredentials([usernamePassword(
            credentialsId: 'dockerhub-creds',
            usernameVariable: 'DOCKER_USER',
            passwordVariable: 'DOCKER_PASS'
        )]) {
            bat """
            echo %DOCKER_PASS% | docker login -u %DOCKER_USER% --password-stdin
            docker push %DOCKER_IMAGE%
            """
```

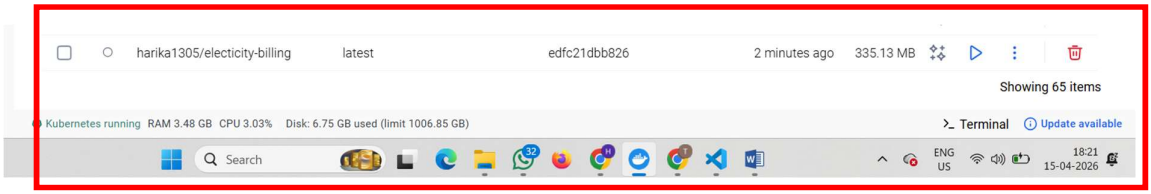
```
            }
        }
    }
    stage('Deploy to Kubernetes') {
        steps {
            withCredentials([file(credentialsId: 'kubeconfig', variable: 'KUBECONFIG')]) {
                bat """
                set KUBECONFIG=%KUBECONFIG%
                kubectl get nodes
                kubectl apply -f deployment.yaml
                """
            }
        }
    }
  }
  post {
    success {
        echo 'CI/CD Pipeline executed successfully!'
    }
    failure {
        echo 'Pipeline failed. Please check logs.'
    }
  }
}
```

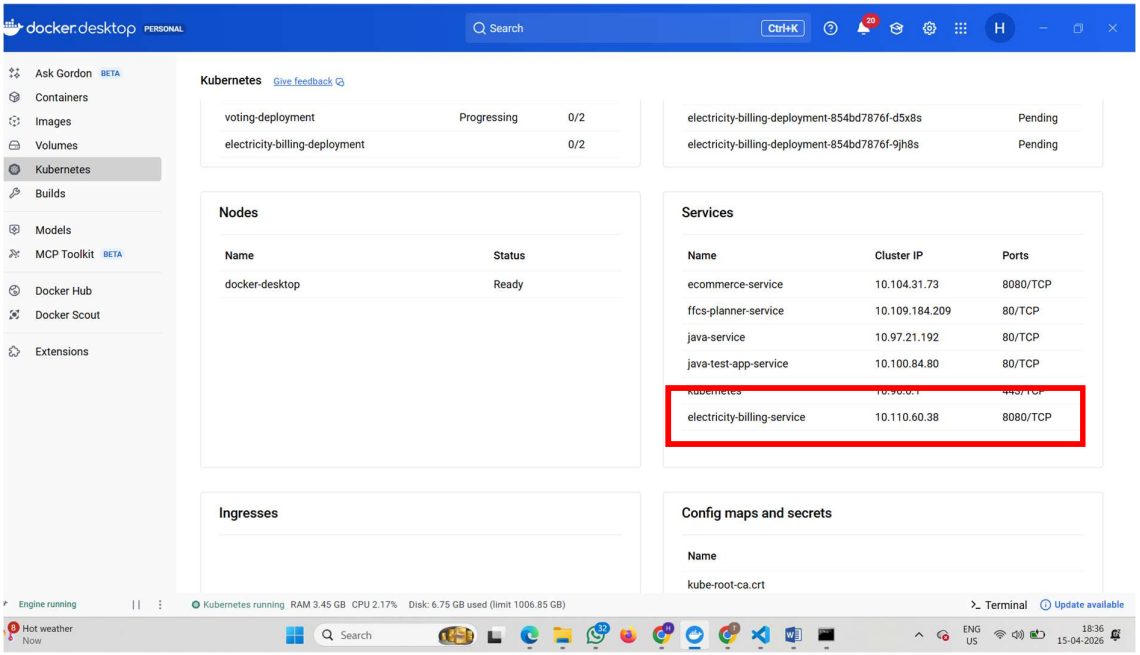


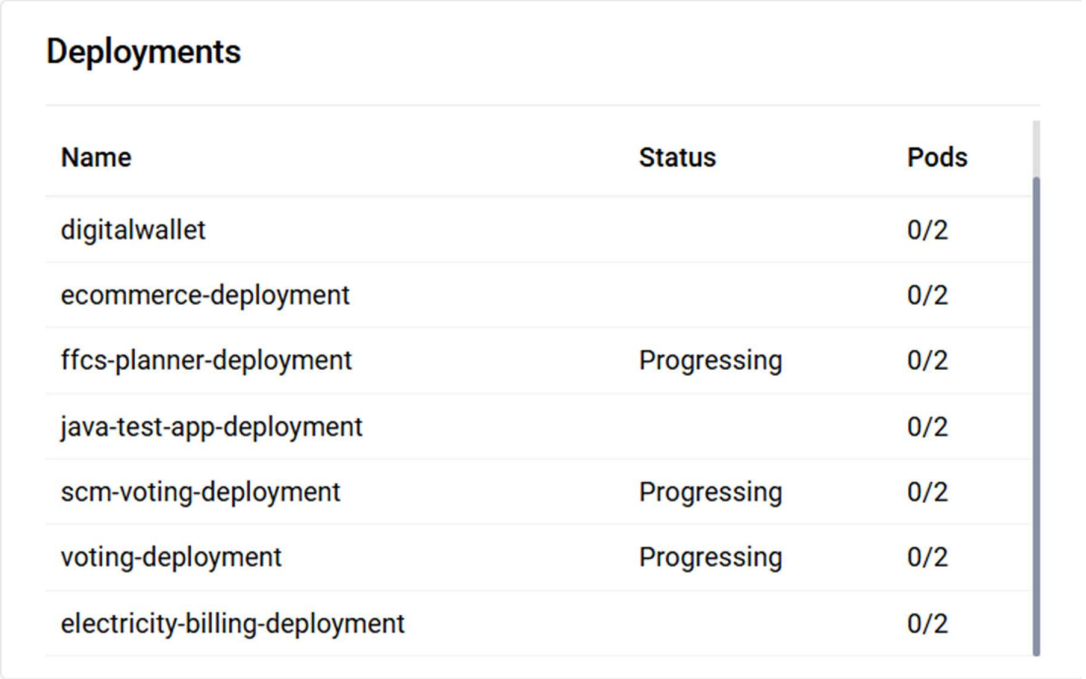
Task-4)

<u>Step 11 </u>[Deploy service on docker and kubernetes]

Verify Docker and **Kubernetes Deployment**







Deployments

Name	Status	Pods
digitalwallet		0/2
ecommerce-deployment		0/2
ffcs-planner-deployment	Progressing	0/2
java-test-app-deployment		0/2
scm-voting-deployment	Progressing	0/2
voting-deployment	Progressing	0/2
electricity-billing-deployment		0/2

```
C:\Users\Lenovo>kubectl get svc
NAME                          TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
ecommerce-service             NodePort    10.104.31.73     <none>        8080:30082/TCP   111m
electricity-billing-service   NodePort    10.110.60.38     <none>        8080:30083/TCP   17m
ffcs-planner-service          NodePort    10.109.184.209   <none>        80:30008/TCP     13d
java-service                  NodePort    10.97.21.192     <none>        80:30010/TCP     12d
java-test-app-service         NodePort    10.100.84.80     <none>        80:30007/TCP     14d
kubernetes                    ClusterIP   10.96.0.1        <none>        443/TCP          155d

C:\Users\Lenovo>
```